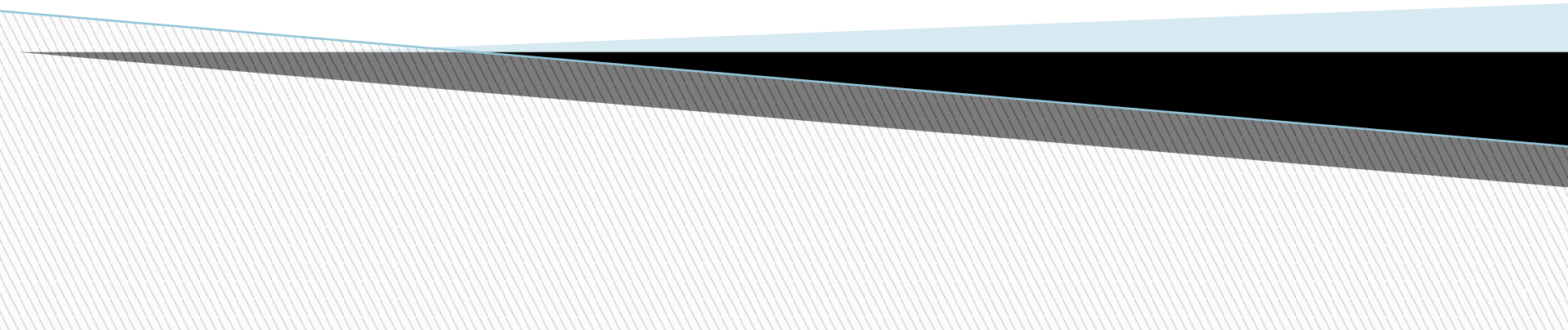


Database Systems (CS-151)

Course Instructor: Munazza



Course administration

Instructor: Munazza

(Lecturer, Deptt of Computer Science, University of Wah, Wah Cantt)

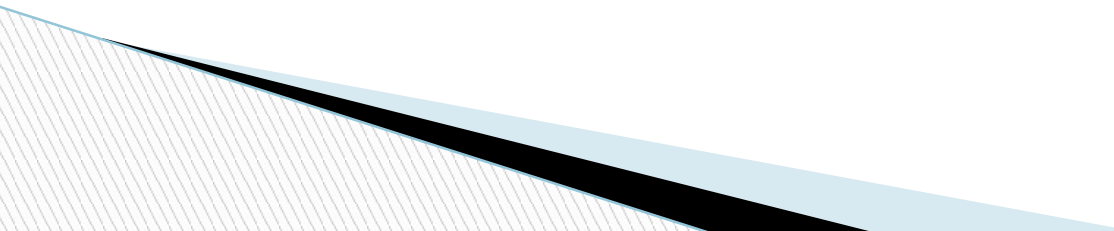
Contact Information:

munazza.bibi@uow.edu.pk

For queries related to course(Assignments/Projects) just leave an email

at **munazza.bibi@uow.edu.pk**

Consultation Hours 02:00PM - 03:00 PM

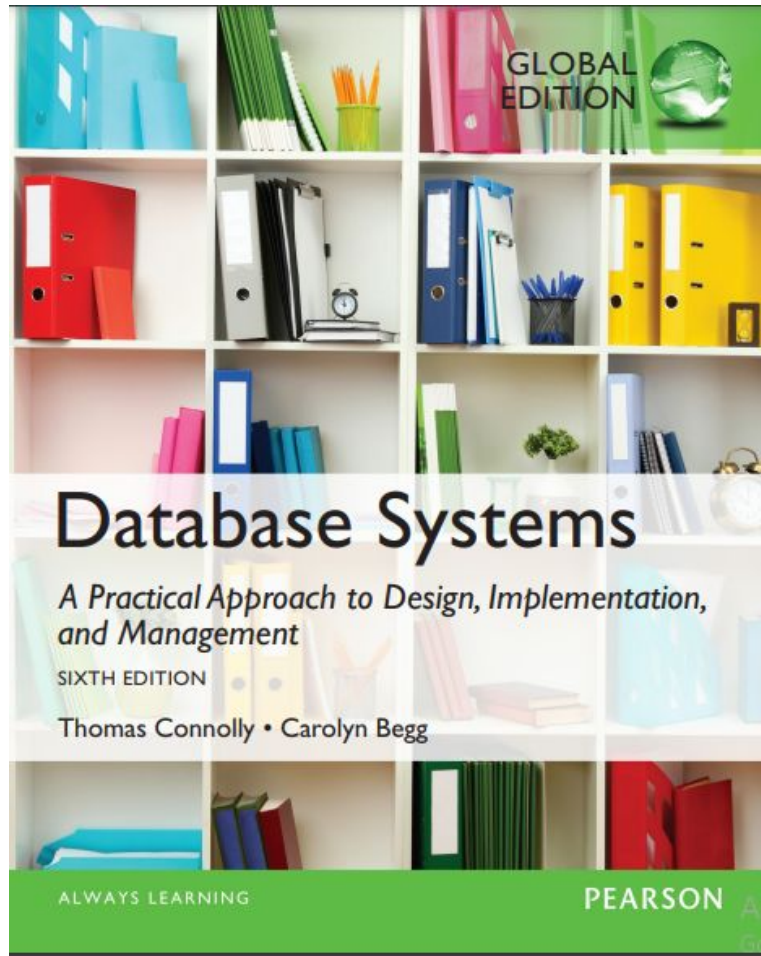


Course Learning Outcomes

Course Learning Outcomes		
After successful completion of this course, the students should be able to:	GA	BT Level
1. Explain fundamental database concepts	2	C2
2. Identify functional dependencies and Produce SQL queries for database definition and manipulation in any DBMS	3	C3
3. Discover conceptual, logical and physical database schemas using different data models	4	C4

Mapping of GA's/PLO's with CLO's

Mapping of CLOs to GA's					
GA's/CLOs	CLO1	CLO2	CLO3	CLO4	
GA1 (Academic Education)					
GA2 (Knowledge for Solving Computing Problems)	✓				
GA3 (Problem Analysis)			✓		
GA4 (Design/Development of Solutions)		✓			
GA5 (Modern Tool Usage)					
GA6 (Individual and Teamwork)					
GA7 (Communication)					
GA8 (Computing Professionalism and Society)					
GA9 (Ethics)					
GA10 (Life-long Learning)					



Recommended Textbook and Reference Books

Text Book:

Database Systems: A Practical Approach to Design, Implementation, and Management by Thomas Connolly and Carolyn Begg 6th Edition, 2021

Text Book:

Database Management Systems by Raghu Ramakrishnan, Johannes Gehrke, 3rd Edition 2002 Database System Concepts by Avi Silberschatz, Henry F. Korth and S. Sudarshan, 7th Edition, 2019

Database Systems: The Complete Book by Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Wido, 2nd Edition, 2008

Goals for today

- ◆ To introduce:
 - **Database Application**
 - **Levels of Data**
 - **Users of Database Systems**
 - **Typical Components of a Database Environment**

Database Application

- ❖ Database Application is a program or group of programs which is used for performing certain operations on the data stored in the database.
- ❖ These operations may contain insertion of data into a database or extracting some data from the database based on a certain condition, updating data in the database, producing the data as output.

Database Management Systems

- ❖ Database management system is a collection of small programs to perform certain operation on data and manage the data. Two basic operations performed by the DBMS are:

Management of Data in the Database

- ❖ Management of the data means to specify that how data will be stored, structured and accessed in the database.

Management of Users associated with the database

- ❖ Management of database users means to manage the users in such a way that they can perform any desired operations on the database.

- ❖ DBMS also ensures that a user can not perform any operation for which he is not allowed. And also an authorized user is not allowed to perform any action which is restricted to that user.
- ❖ In General DBMS is a collection of Programs performing all necessary actions associated to a database

Levels of Data

1) Real World Data:

The real world level of data means that level of data at which entities or objects exist in reality, it means that any object existing in reality have a name and other identifiable attributes through which we can identify that specific object or entity. Example: Any Student

Levels of Data

2) Meta Data:

Meta Data is data about data. It tells that what type of data will be stored in the database, what will be size of a certain attribute of the real world data, how many and what attributes will be used to store the data about the entity in the database.

Example: Name ,Character Type, 25 character size field,
 Age, Date type, 8 bytes size
 Class, Alpha Numeric, 8 byte size field

Levels of Data

3) Existence of Data:

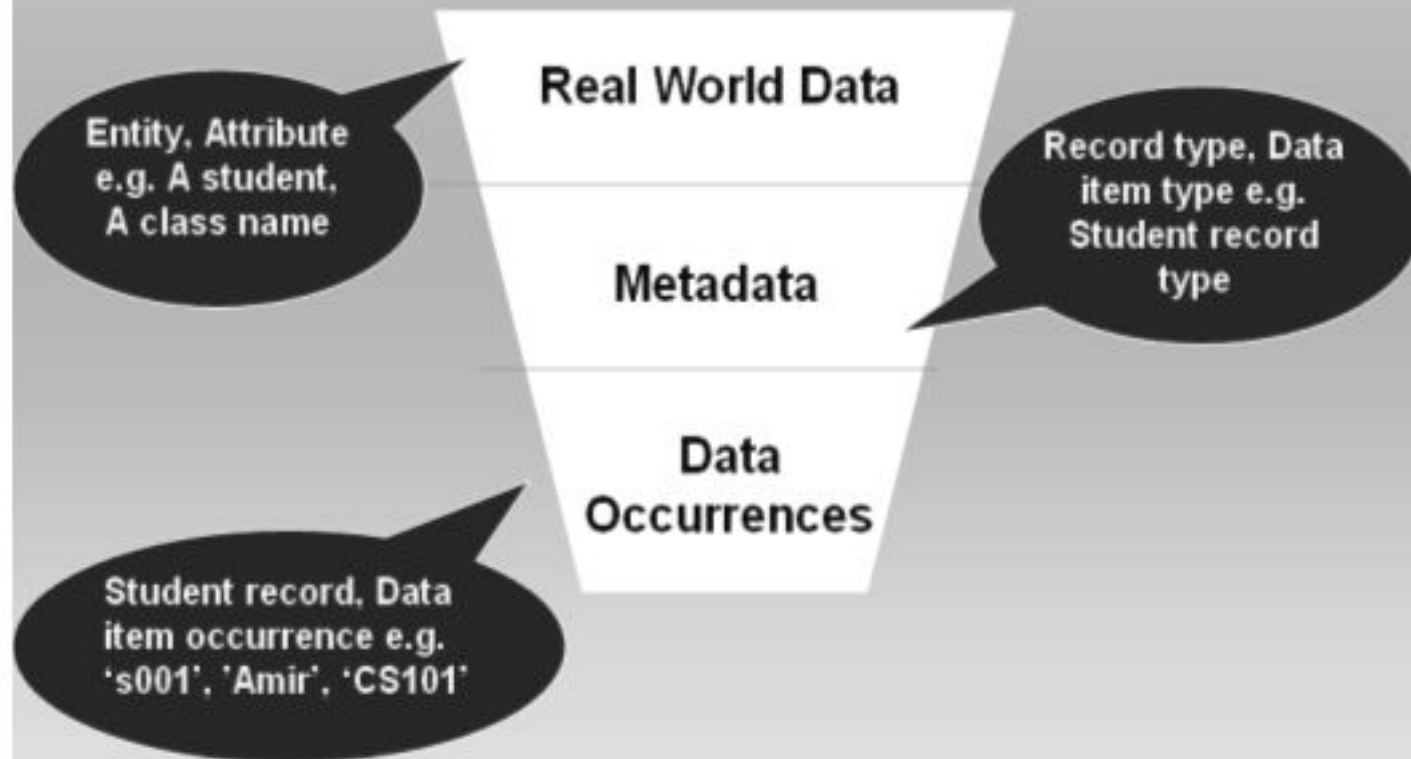
Existence of the data level shows the actual data regarding the entities as real world level according to the rules define at the Meta Data level.

Example:

According to the definition given in the Meta data level the Actual data or Data occurrence for the entity at real world level is shown below:

Name	Age	Class	
Ali	20/8/1979	MCS-I	
Amir	22/3/1978	MCS-II	etc...

Levels of Data



Users of Database Systems

◆ Users of Database Systems:

1. Application Programmers
2. End Users
 - Naïve
 - Sophisticated

Users of Database Systems

Application programmers:

- ❖ This category of database users contains those people who create different types of database application programs.
- ❖ Application programmers design the application according to the needs of the other users of the database in a certain environment.
- ❖ Application programmers are skilled people who have clear idea of the structure of the database and know clearly about the needs of the organizations.

Users of Database Systems

End Users

- ❖ Second category of the Database users are the end users, this group of users contains the people who use the database application programs developed by the Application programmers. This category further contains two types of users:

- (i) Naïve Users

- (ii) Sophisticated Users

Users of Database Systems

Naïve Users

- ❖ these are the users who use the existing application to interact with the database. For example, online library system, ticket booking systems, ATMs etc which has existing application and users use them to interact with the database to fulfill their requests.

Users of Database Systems

Sophisticated Users

- ❖ They are database developers, who write SQL queries to select/insert/delete/update data. They do not use any application or programs to request the database. They directly interact with the database by means of query language like SQL.

Users of Database Systems

- ◆ **Database Administrators (DBA):** The administration and maintenance of database is taken care by database Administrator – DBA.
- ◆ DBA is a very responsible position in an organization. He is responsible for proper working of the database and DBMS, has the responsibility of making proper database backups and make necessary actions for recovering the database in case of a database crash.
- ◆ To fulfill the requirements of a DBA position a DBA needs vast experience and very elegant technical skills.

Users of Database Systems

Duties of the DBA

- ❖ Installing and upgrading the DBMS
- ❖ Granting data access
- ❖ Routine Maintenance
- ❖ Backups
- ❖ Monitoring disk space
- ❖ Monitoring jobs running

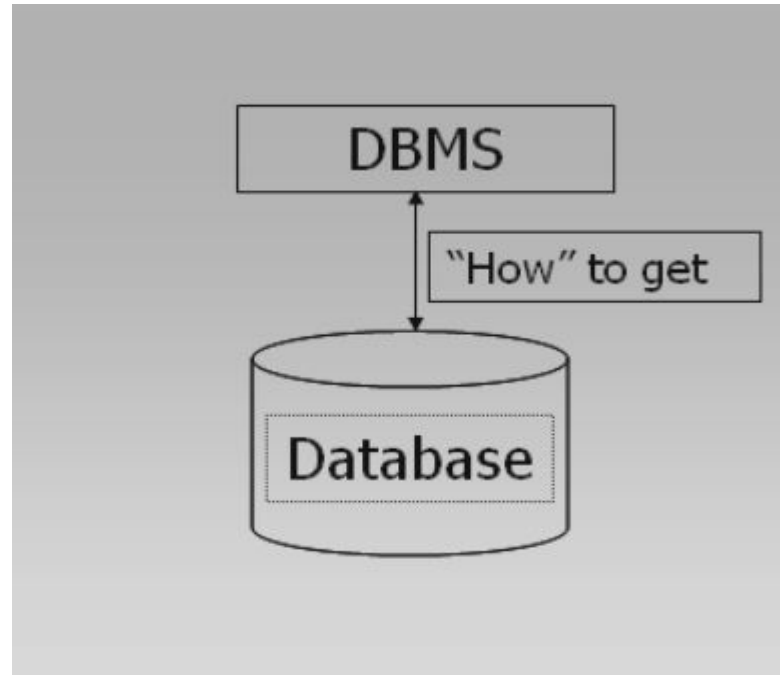
Duties of DBA

- ◆ **Installing and upgrading the DBMS:** DBA is responsible for installing a new DBMS server for the new projects. He is also responsible for upgrading these servers as new versions come in the market.
- ◆ **Granting Access to Users:** DBA is also responsible for grant of access rights to the database users. Along with granting and revoking (taking back) the rights the DBA continuously monitors and ensure the legal use of these rights.

Duties of DBA

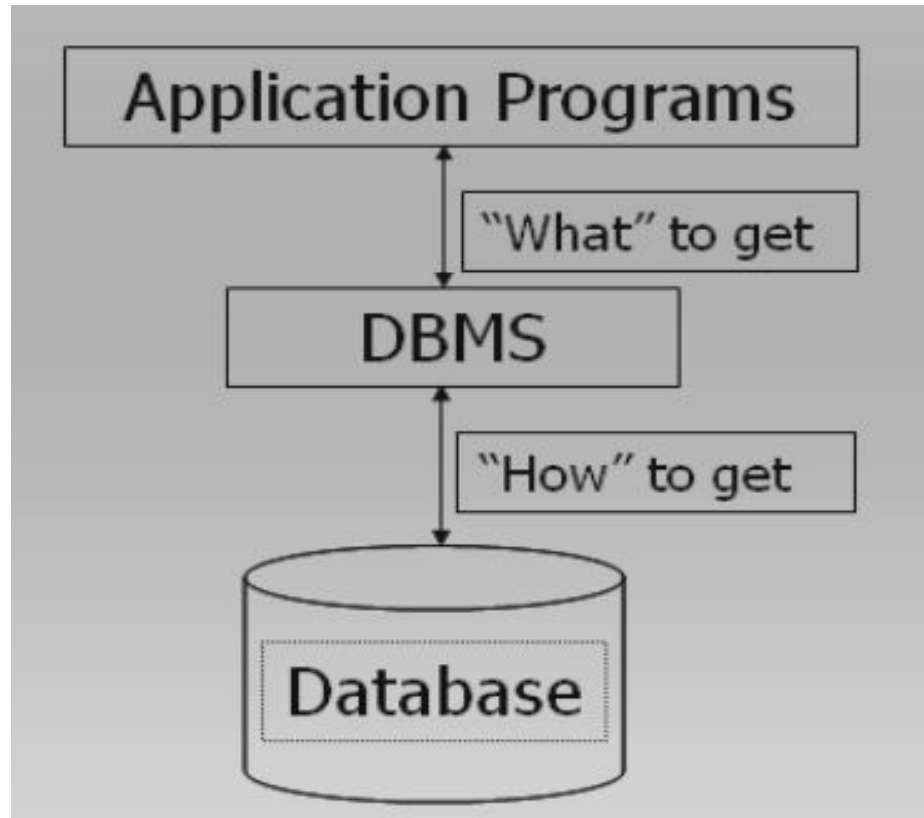
- ◆ **Monitoring Disk Space :** When a new database is created it takes a limited space but as a result of daily activity the database acquires more data and grows in size very rapidly. The DBA has to monitor the disk space usage and statistics to ensure that no data over flow occurs at any stage.
- ◆ **Monitoring Running Jobs:** To ensure the secure and proper functioning of the database system a DBA continuously monitors some associated activities and ensure that all users are using their authorities legally and different devices attached to the database system are functioning properly.

Typical Components of a Database Environment



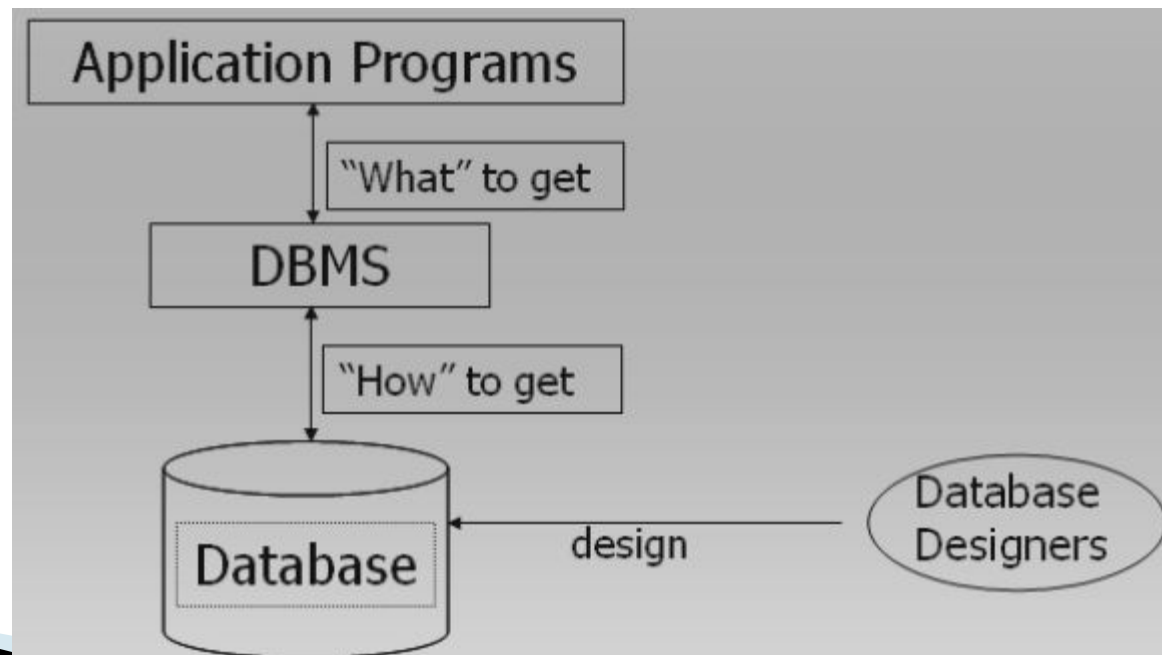
Database is used to store data and DBMS uses mechanisms to get data from the database

Application Programs



Application Programs talk to DBMS and ask for data

- ❖ Database designers design the database (for large organizations) and install the DBMS for use by the users of the database in any specific organization.



Database Administrator

